

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Branch: CSE-AIML

Experiment No.: 3

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1. TITLE

Implementation of JavaScript Arrays and Functions using Node.js

2. AIM

To write and execute a JavaScript program using **arrays and functions** in Node.js, demonstrating array creation, traversal, searching, calculation, and manipulation through user-defined functions.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the use of arrays in JavaScript.
 - To create and manipulate arrays using built-in array operations.
 - To define and invoke user-defined functions.
 - To pass arrays as function arguments.
 - To perform searching and calculations on array elements.
 - To execute JavaScript programs using the Node.js runtime.
 - To understand how functions can be used to organize reusable program logic.
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4. THEORY

JavaScript Arrays

An array is a data structure used to store multiple values in a single variable. JavaScript arrays can contain values of different data types and are dynamically sized.

Example:

```
let numbers = [10, 20, 30, 40, 50];
```

Array elements are accessed using their index, beginning from 0.

```
console.log(numbers[0]);
```

Output:

10

Common array operations include:

- `push()` – adds an element to the end.
- `pop()` – removes the last element.
- `shift()` – removes the first element.
- `unshift()` – adds an element to the beginning.
- `includes()` – checks whether an element exists.
- `indexOf()` – returns the index of an element.
- `length` – returns the number of elements.

JavaScript Functions

A function is a reusable block of code designed to perform a particular task.

Example:

```
function add(a, b) {  
    return a + b;  
}
```

The function can be invoked as:

```
console.log(add(10, 20));
```

Functions can accept arrays as parameters and return calculated or processed results.

Node.js Execution

Node.js allows JavaScript programs to execute outside a web browser.

The program can be saved as:

array_functions.js

and executed from the terminal using:

node array_functions.js

5. ALGORITHM / PROCEDURE

1. Create a JavaScript file named `array_functions.js`.
 2. Create an array containing a set of numbers.
 3. Define a function to display all elements of the array.
 4. Define a function to calculate the sum of the array elements.
 5. Define a function to calculate the average.
 6. Define a function to find the largest element.
 7. Define a function to find the smallest element.
 8. Define a function to search for an element in the array.
 9. Define a function to create a new array containing only even numbers.
 10. Invoke all the functions with appropriate test data.
 11. Display the results using `console.log()`.
 12. Execute the program using Node.js.
 13. Verify the output for all test cases.
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6. PROGRAM

`array_functions.js`

```
// JavaScript Program: Arrays and Functions
```

```
// Execution Environment: Node.js
```

```
// Function to display all elements of an array
```

```
function displayArray(arr) {  
    console.log("Array Elements:", arr.join(" "));  
}
```

```
// Function to calculate the sum of array elements
```

```
function calculateSum(arr) {  
    let sum = 0;  
  
    for (let i = 0; i < arr.length; i++) {  
        sum += arr[i];  
    }  
  
    return sum;  
}  
  
// Function to calculate the average  
function calculateAverage(arr) {  
    if (arr.length === 0) {  
        return 0;  
    }  
  
    return calculateSum(arr) / arr.length;  
}  
  
// Function to find the largest element  
function findLargest(arr) {  
    if (arr.length === 0) {  
        return undefined;  
    }  
  
    let largest = arr[0];
```

```
for (let i = 1; i < arr.length; i++) {  
    if (arr[i] > largest) {  
        largest = arr[i];  
    }  
}  
  
return largest;  
}  
  
// Function to find the smallest element  
function findSmallest(arr) {  
    if (arr.length === 0) {  
        return undefined;  
    }  
  
    let smallest = arr[0];  
  
    for (let i = 1; i < arr.length; i++) {  
        if (arr[i] < smallest) {  
            smallest = arr[i];  
        }  
    }  
  
    return smallest;  
}
```

// Function to search for an element

```
function searchElement(arr, element) {  
    return arr.includes(element);  
}
```

// Function to return only even numbers

```
function getEvenNumbers(arr) {  
    return arr.filter(function (number) {  
        return number % 2 === 0;  
    });  
}
```

// Function to add an element to the array

```
function addElement(arr, element) {  
    arr.push(element);  
    return arr;  
}
```

// Function to remove the last element

```
function removeLastElement(arr) {  
    arr.pop();  
    return arr;  
}
```

```
// -----
```

```
// Test Data
```

```
// -----
```

```
let numbers = [10, 25, 30, 45, 50];
```

```
console.log("==== JavaScript Arrays and Functions =====");
```

```
console.log();
```

```
// Test 1: Display array
```

```
console.log("1. Display Array");
```

```
displayArray(numbers);
```

```
console.log();
```

```
// Test 2: Sum
```

```
console.log("2. Sum of Array Elements");
```

```
console.log("Sum =", calculateSum(numbers));
```

```
console.log();
```

```
// Test 3: Average
```

```
console.log("3. Average of Array Elements");
```

```
console.log("Average =", calculateAverage(numbers));
```

```
console.log();
```

// Test 4: Largest element

```
console.log("4. Largest Element");
```

```
console.log("Largest =", findLargest(numbers));
```

```
console.log();
```

// Test 5: Smallest element

```
console.log("5. Smallest Element");
```

```
console.log("Smallest =", findSmallest(numbers));
```

```
console.log();
```

// Test 6: Search existing element

```
console.log("6. Search for Existing Element (30)");
```

```
console.log("Found =", searchElement(numbers, 30));
```

```
console.log();
```

// Test 7: Search non-existing element

```
console.log("7. Search for Non-existing Element (100)");
```

```
console.log("Found =", searchElement(numbers, 100));
```

```
console.log();
```



```
// Test 8: Even numbers

console.log("8. Even Numbers");

console.log("Even Numbers =", getEvenNumbers(numbers));

console.log();
```

```
// Test 9: Add an element

console.log("9. Add Element (60)");

let updatedArray = addElement([...numbers], 60);

displayArray(updatedArray);

console.log();
```

```
// Test 10: Remove last element

console.log("10. Remove Last Element");

let modifiedArray = removeLastElement([...numbers]);

displayArray(modifiedArray);
```

7. CODE EXPLANATION

Code / Concept	Explanation
<code>let numbers = [...]</code>	Creates an array containing numerical values.
<code>function displayArray(arr)</code>	Defines a function that accepts an array as an argument.

<code>arr.join(" ")</code>	Converts array elements into a space-separated string for display.
<code>calculateSum(arr)</code>	Calculates the sum of all elements using a <code>for</code> loop.
<code>calculateAverage(arr)</code>	Calculates the average using the sum and array length.
<code>findLargest(arr)</code>	Iterates through the array to find the largest value.
<code>findSmallest(arr)</code>	Iterates through the array to find the smallest value.
<code>arr.includes(element)</code>	Checks whether the specified element exists in the array.
<code>arr.filter()</code>	Creates a new array containing elements satisfying a condition.
<code>arr.push(element)</code>	Adds an element to the end of an array.
<code>arr.pop()</code>	Removes the last element from an array.
<code>[...numbers]</code>	Creates a copy of the original array using the spread operator.
<code>console.log()</code>	Displays output in the Node.js terminal.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Status
TC-01	Display [10, 25, 30, 45, 50]	All array elements are displayed	Pass
TC-02	Calculate sum	Sum = 160	Pass
TC-03	Calculate average	Average = 32	Pass
TC-04	Find largest element	Largest = 50	Pass
TC-05	Find smallest element	Smallest = 10	Pass
TC-06	Search for 30	Element found → true	Pass
TC-07	Search for 100	Element not found → false	Pass
TC-08	Extract even numbers	[10, 30, 50]	Pass
TC-09	Add 60	[10, 25, 30, 45, 50, 60]	Pass
TC-10	Remove last element	[10, 25, 30, 45]	Pass

9. OUTPUT

Execution Command

node array_functions.js

Output

===== JavaScript Arrays and Functions =====

1. Display Array

Array Elements: 10 25 30 45 50

2. Sum of Array Elements

Sum = 160

3. Average of Array Elements

Average = 32

4. Largest Element

Largest = 50

5. Smallest Element

Smallest = 10

6. Search for Existing Element (30)

Found = true

7. Search for Non-existing Element (100)

Found = false

8. Even Numbers

Even Numbers = [10, 30, 50]

9. Add Element (60)

Array Elements: 10 25 30 45 50 60

10. Remove Last Element

Array Elements: 10 25 30 45

10. RESULT

Thus, the JavaScript program was successfully implemented and executed using **Node.js**. The experiment demonstrated the use of **arrays, user-defined functions, array traversal, searching, filtering, and array manipulation**, with all test cases producing the expected results.